

Combinatorial interpretations in affine Coxeter groups of types B, C, and D

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What is a Coxeter group?

A **Coxeter group** is a group with

- ▶ **Generators:** $\{s_1, s_2, \dots, s_n\}$
- ▶ **Relations:** $s_i^2 = 1$, $(s_i s_j)^{m_{ij}} = 1$ where $m_{ij} \geq 2$ or $= \infty$

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 - ▶ $m_{ij} = 2$: $(s_i s_j)(s_i s_j) = 1 \implies s_i s_j = s_j s_i$ (they commute)
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 - ▶ $m_{ij} = \infty$: s_i and s_j are not related.

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- ▶ They're awesome.

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Why Coxeter groups?

- ▶ They're awesome.
- ▶ Discrete Geometry: Symmetries of regular polyhedra.
- ▶ Algebra: Symmetric group generalizations. (Kac-Moody, Hecke)
- ▶ Geometry: Classification of Lie groups and Lie algebras

Examples of Coxeter groups

(Finite) n -Permutations S_n (A_{n-1})

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231	312
321	321

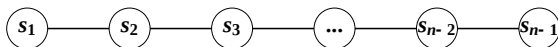
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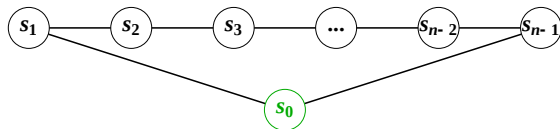


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Affine n -Permutations $\tilde{S}_n$ ($\tilde{A}_{n-1}$)

- Generators: $\{s_0, s_1, \dots, s_{n-1}\}$
- S_n is a parabolic subgroup of $\tilde{S}_n$

Minimal length coset representatives

For a Coxeter group $\widetilde{W}$ generated by $\{s_0, s_1, \dots, s_n\}$,

- ▶ An induced subgraph of $\widetilde{W}$'s Coxeter graph is a subgroup W
 - ▶ Today, we will always have W defined by $\widetilde{W} \setminus \{s_0\}$

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Simple example: For $\tilde{w} = s_0 s_1 s_2 s_3 s_0 s_1 s_2 \in \widetilde{S}_4$
 $\tilde{w} = s_0 s_1 s_2 s_3 s_0 s_1 s_2$

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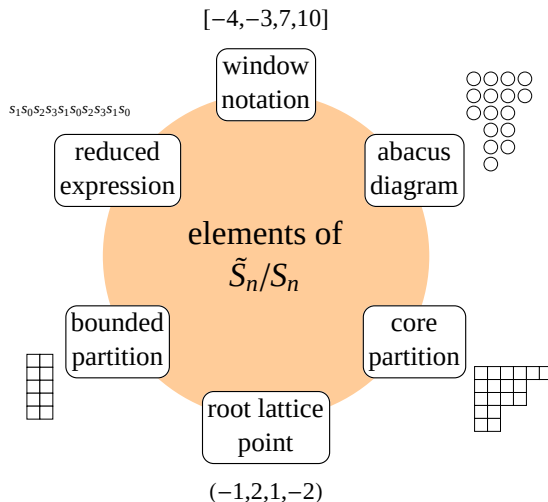
Less simple example: $\tilde{w} = s_1 s_3 s_2 s_3 s_0 s_1 \in \widetilde{S}_4$

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★ Combinatorial interpretations are easier to use. ★

Combinatorial interpretations of $\tilde{S}_n/S_n$



Window notation

Affine n -Permutations $\tilde{S}_n$ (G. Lusztig 1983, H. Eriksson, 1994)

Write an element $\tilde{w} \in \tilde{S}_n$ in 1-line notation as a permutation of $\mathbb{Z}$.

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Generators transpose **infinitely many** pairs of entries:

$$s_i : (\mathbf{i}) \leftrightarrow (\mathbf{i+1}) \dots (n+i) \leftrightarrow (n+i+1) \dots (-n+i) \leftrightarrow (-n+i+1) \dots$$

In $\tilde{S}_4$,	$\dots w(-4)$	$w(-3) \ w(-2) \ w(-1) \ w(0)$	$w(1) \ w(2) \ w(3) \ w(4)$	$w(5) \ w(6) \ w(7) \ w(8)$	$w(9) \dots$
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★ For $\tilde{w} = w^0 w$, the window of w^0 is the window of $\tilde{w}$, **sorted** ↗.

An abacus model for $\tilde{S}_n/S_n$

(James and Kerber, 1981) Given $w^0 = [w_1, \dots, w_n] \in \tilde{S}_n/S_n$,

- Place integers in n *runners*.

-15 -14 -13 -12

-11 -10 -9 -8

-7 -6 -5 -4

-3 -2 -1 0

1 2 3 4

5 6 7 8

9 10 11 12

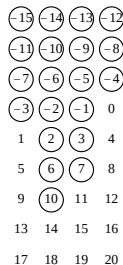
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- ▶ **Bijection:** Given w^0 , create an abacus where each runner has a lowest bead at w_i .

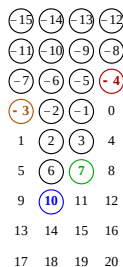
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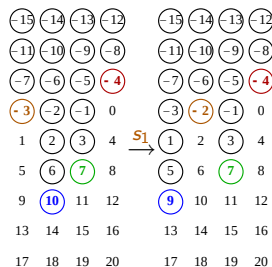
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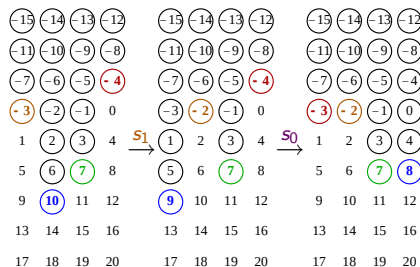
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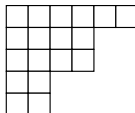
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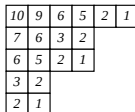
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Integer partitions and n -core partitions

For an integer partition $\lambda = (\lambda_1, \dots, \lambda_k)$ drawn as a Ferrers diagram,

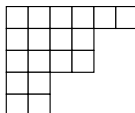


The *hook length* of a box is # boxes below and to the right.

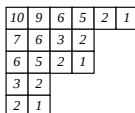


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The *hook length* of a box is # boxes below and to the right.



An n -core is a partition with no boxes of hook length dividing n .

Example. λ is a 4-core, 8-core, 11-core, 12-core, etc.

λ is NOT a 1-, 2-, 3-, 5-, 6-, 7-, 9-, or 10-core.

Core partitions for $\tilde{S}_n/S_n$

Elements of $\tilde{S}_n/S_n$ are in bijection with n -cores.

Bijection: $\{\text{abaci}\} \longleftrightarrow \{n\text{-cores}\}$

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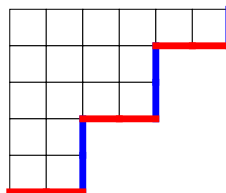
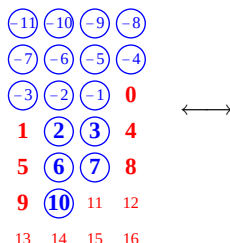
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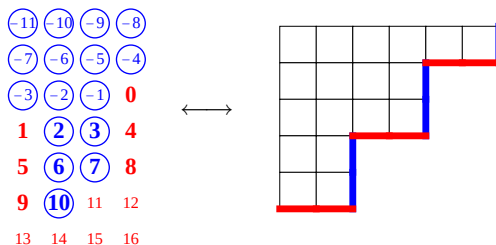
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Fact: Abacus flush with n -runners $\leftrightarrow$ partition is n -core.

Action of generators on the core partition

- Label the boxes of λ with residues.
- s_i acts by adding or removing boxes with residue i .

0	1	2	3	0	1
3	0	1	2	3	0
2	3	0	1	2	3
1	2	3	0	1	2
0	1	2	3	0	1
3	0	1	2	3	0

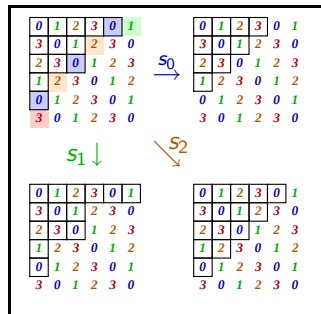
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Example. $\lambda = (5, 3, 3, 1, 1)$

- ▶ has removable 0 boxes (s_0 is a descent)
- ▶ has addable 1, 2, 3 boxes (s_1, s_2, s_3 are ascents)

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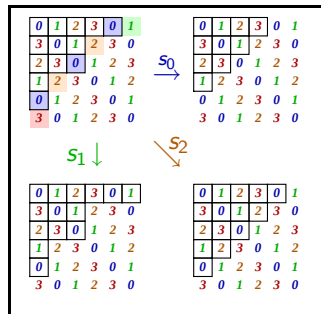
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Idea: Use to determine a canonical reduced expression for w^0 .

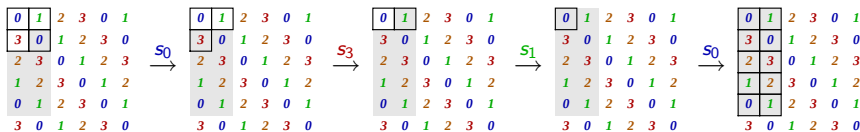
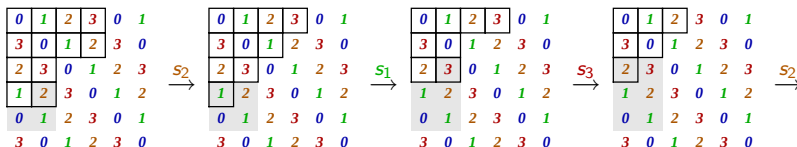
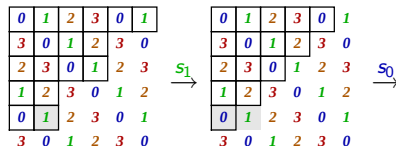
- ▶ Tally residues from bottom to top, from right to left.



Canonical reduced expression for $\tilde{S}_n/S_n$

Example: Reduced expression corresponding to $\lambda = (6, 4, 4, 2, 2)$:

$$\mathcal{R}(\lambda) = s_1 s_0 s_2 s_1 s_3 s_2 s_0 s_3 s_1 s_0$$



Bounded partitions for $\tilde{S}_n/S_n$

A partition $\beta = (\beta_1, \dots, \beta_k)$ is *b -bounded* if $\beta_i \leq b$ for all i .

Elements of $\tilde{S}_n/S_n$ are in bijection with $(n-1)$ -bounded partitions.

Bijection: (Lapointe, Morse, 2005)

$$\{n\text{-cores } \lambda\} \leftrightarrow \{(n-1)\text{-bounded partitions } \beta\}$$

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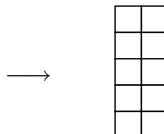
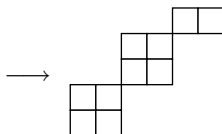
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- ▶ Remove all boxes of λ with hook length $\geq n$
- ▶ Left-justify remaining boxes.

10	9	6	5	2	1
7	6	3	2		
6	5	2	1		
3	2				
2	1				

$$\lambda = (6, 4, 4, 2, 2)$$



$$\beta = (2, 2, 2, 2, 2)$$

Canonical reduced expression for $\tilde{S}_n/S_n$

Given the bounded partition, read off the reduced expression:

Method: (Berg, Jones, Vazirani, 2009)

- ▶ Fill β with residues i
- ▶ Tally s_i reading right-to-left in rows from bottom-to-top

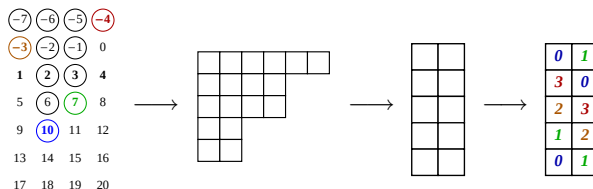
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- Fill β with residues i
- Tally s_i reading right-to-left in rows from bottom-to-top

Example. $[-4, -3, 7, 10] = s_1 s_0 s_2 s_1 s_3 s_2 s_0 s_3 s_1 s_0.$



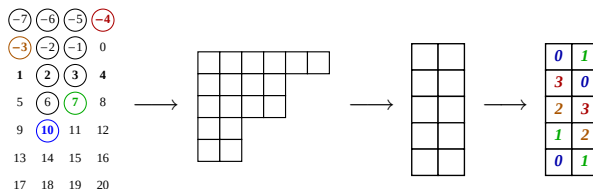
Canonical reduced expression for $\tilde{S}_n/S_n$

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- The Coxeter length of w^0 is the number of boxes in β .

Summary for $\tilde{S}_n/S_n$

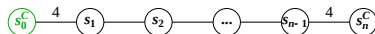
- ▶ See S_n as parabolic subgroup of $\tilde{S}_n$
- ▶ **Window notation**
 - ▶ $\tilde{S}_n$ elements can be written as a permutation of $\mathbb{Z}$
 - ▶ Min. len. coset rep's correspond to **sorted** $\mathbb{Z}$ -permutations.
- ▶ **Abacus models**
 - ▶ Define the abacus by reading the entries from a window
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- ▶ **Core partitions**
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Summary for $\widetilde{W}/W$

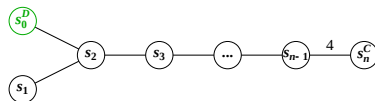
- ▶ See W as parabolic subgroup of $\widetilde{W}$
- ▶ **Window notation**
 - ▶ $\widetilde{W}$ elements can be written as a permutation of $\mathbb{Z}$
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- ▶ **Abacus models**
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Coxeter Graphs for Types $\tilde{B}$, $\tilde{C}$, $\tilde{D}$

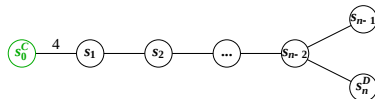
Type $\tilde{C}/C$:



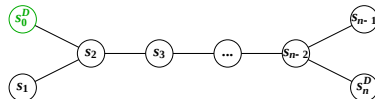
Type $\tilde{B}/B$:



Type $\tilde{B}/D$:



Type $\tilde{D}/D$:



Window notation for $\widetilde{W}$

Write $\widetilde{w} \in \widetilde{W}$ as a **mirrored permutation of $\mathbb{Z}$** with period $N = 2n + 1$.

- Satisfies $\widetilde{w}(i + N) = \widetilde{w}(i) + N$ and $\widetilde{w}(-i) = -\widetilde{w}(i)$.

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 - ▶ s_i : switch $\widetilde{w}(i) \leftrightarrow \widetilde{w}(i + 1)$
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In $\widetilde{C}_4$	...	$w(-4)$	$w(-3)$	$w(-2)$	$w(-1)$	$w(0)$	$w(1)$	$w(2)$	$w(3)$	$w(4)$	$w(5)$	$w(6)$	$w(7)$	$w(8)$	$w(9)$	$w(10)$	...
s_1	...	-4	-3	-1	-2	0	2	1	3	4	5	6	8	7	9	11	...
s_0	...	-4	-3	-2	1	0	-1	2	3	4	5	6	7	10	9	8	...
s_4	...	-5	-3	-2	-1	0	1	2	3	5	4	6	7	8	9	10	...

Window notation for $\widetilde{W}$

Write $\widetilde{w} \in \widetilde{W}$ as a **mirrored permutation of $\mathbb{Z}$** with period $N = 2n + 1$.

- ▶ Satisfies $\widetilde{w}(i + N) = \widetilde{w}(i) + N$ and $\widetilde{w}(-i) = -\widetilde{w}(i)$.
- ▶ Define action of generators on $[\widetilde{w}(1), \widetilde{w}(2), \dots, \widetilde{w}(n)]$; extend:
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In $\widetilde{C}_4$	...	$w(-4)$	$w(-3)$	$w(-2)$	$w(-1)$	$w(0)$	$w(1)$	$w(2)$	$w(3)$	$w(4)$	$w(5)$	$w(6)$	$w(7)$	$w(8)$	$w(9)$	$w(10)$	...
s_1	...	-4	-3	-1	-2	0	2	1	3	4	5	6	8	7	9	11	...
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$s_1 s_0$	...	-4	-3	-1	2	0	-2	1	3	4	5	6	8	11	9	7	...

$\widetilde{w}$ is defined by the **window** $[\widetilde{w}(1), \widetilde{w}(2), \dots, \widetilde{w}(n)]$. $s_1 s_0 = [-2, 1, 3, 4]$

Window notation for $\widetilde{W}$

In type $\widetilde{B}$ (version 1) and type $\widetilde{D}$, the s_0 generator is:

- ▶ s_0^D : switch $\{\widetilde{w}(-2), \widetilde{w}(-1)\} \leftrightarrow \{\widetilde{w}(1), \widetilde{w}(2)\}$
- ▶ Therefore, $|\{i < 0 : w(i) > 0\}|$ is even.

In $\widetilde{D}_4$	...	$w(-4)$	$w(-3)$	$w(-2)$	$w(-1)$	$w(0)$	$w(1)$	$w(2)$	$w(3)$	$w(4)$	$w(5)$	$w(6)$	$w(7)$	$w(8)$	$w(9)$	$w(10)$	...
s_0^D	...	-4	-3	1	2	0	-2	-1	3	4	5	6	10	11	9	7	...
s_4^D	...	-6	-5	-2	-1	0	1	2	5	6	3	4	7	8	9	10	...

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s_4^D	...	-6	-5	-2	-1	0	1	2	5	6	3	4	7	8	9	10	...

In type $\widetilde{B}$ (version 2) and type $\widetilde{D}$, the s_n generator is:

- ▶ s_n^D : switch $\{\widetilde{w}(n-1), \widetilde{w}(n)\} \leftrightarrow \{\widetilde{w}(n+1), \widetilde{w}(n+2)\}$
- ▶ Therefore, $|\{i \geq n+1 : w(i) \leq n\}|$ is even.

Window notation for $\widetilde{W}/W$

Theorem. Given an element $\widetilde{w} \in \widetilde{W}$ written as a mirrored permutation of $\mathbb{Z}$, we obtain its minimal length coset representative $w^0 \in \widetilde{W}/W$ by sorting the entries in the base window:

Type	Sorting conditions
$\widetilde{C}/C$	$w(1) < w(2) < \cdots < w(n) < w(n+1)$
$\widetilde{B}/B$	$w(1) < w(2) < \cdots < w(n) < w(n+1)$ Elements of $\widetilde{B}_n/B_n$ are elements of $\widetilde{C}_n/C_n$.
$\widetilde{B}/D$	$w(1) < w(2) < \cdots < w(n) < w(n+2)$ Elements of $\widetilde{B}_n/D_n$ are not necessarily elements of $\widetilde{C}_n/C_n$.
$\widetilde{D}/D$	$w(-2) < w(1) < w(2) < \cdots < w(n) < w(n+2)$ Elements of $\widetilde{D}_n/D_n$ are also elements of $\widetilde{B}_n/D_n$.

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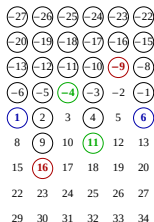
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► It makes sense to define abaci for $\widetilde{W}/W$!

Abacus models for $\widetilde{W}/W$

(Hanusa and Jones, 2011) Given $w^0 = [w_1, \dots, w_{2n}] \in \widetilde{W}/W$, create an abacus with $2n$ runners with lowest beads in positions w_i .

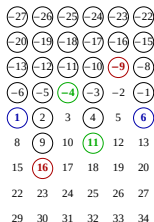
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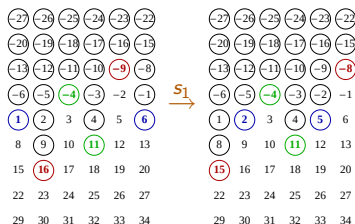
Again, generators interchange runners:

- ▶ $s_i: (i) \leftrightarrow (i+1) \ \& \ (2n-i) \leftrightarrow (2n-i+1).$
- ▶ $s_0^C: 1 \overset{\text{shift}}{\leftrightarrow} 2n$
- ▶ $s_n^C: n \leftrightarrow n+1$
- ▶ $s_0^D: \{1, 2\} \overset{\text{shift}}{\leftrightarrow} \{2n-1, 2n\}$
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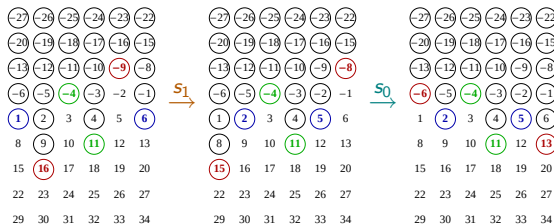
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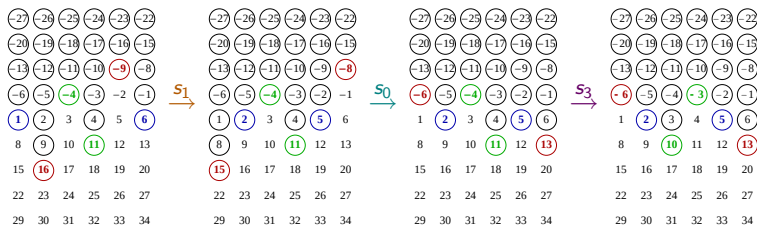
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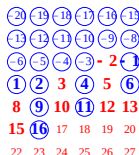
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Structure of abaci and cores in $\widetilde{W}/W$

In abaci:

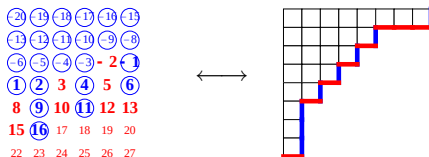
- **Symmetry:** bead in position $i \leftrightarrow$ gap in position $2n + 1 - i$.
- If s_0^D : number of gaps $< 2n + 1$ is even. (**even** abacus)



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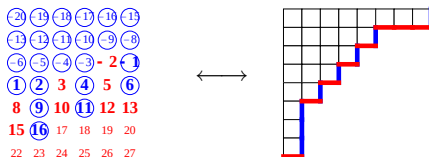
Under the bijection between abaci and core partitions,

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Under the bijection between abaci and core partitions,

- **Symmetry:** Abaci in $\widetilde{W}/W \leftrightarrow$ *Self-conjugate* $(2n)$ -cores
- If s_0^D : even number of boxes on the main diagonal (**even** core)
- Know the action of generators on cores.

Residue Structure in $\widetilde{W}/W$

In $\widetilde{C}_n/C_n$, we have fixed residue structure.

The residues increase from 0 up to n and back down to 0 :

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
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2	1	0	1	2	3	2	1	0	1
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► s_i acts by adding or removing boxes with residue i .

Example: $[-9, -4, 1, 6, 11, 16] = s_1 s_0 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1 s_0$.

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Example: $[-9, -4, 1, 6, 11, 16] = s_1 s_0 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1 s_0$.

Residue Structure in $\widetilde{W}/W$

In $\widetilde{C}_n/C_n$, we have fixed residue structure.

The residues increase from 0 up to n and back down to 0:

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

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3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

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3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

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2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

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1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

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Example: $[-9, -4, 1, 6, 11, 16] = s_1 s_0 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1 s_0$.

Residue Structure in $\widetilde{W}/W$

In other types, residues involving $\{n-1, n\}$ and $\{0, 1\}$ depend on λ .

In type $\widetilde{D}/D$, there are both **escalators** and **descalators**.

0	0	2	3				3	2				2	3
0	0	1	2	3				3	2				2
2	1	0	0	2	3				3	2			
3	2	0	0	1	2	3				3	2		
	3	2	1	0	0	2	3				3	2	
		3	2	0	0	1	2	3				3	2
			3	2	1	0	0	2	3				3
3				3	2	0	0	1	2	3			
2	3				3	2	1	0	0	2	3		
	2	3				3	2	0	0	1	2	3	
		2	3				3	2	1	0	0	2	3
			2	3				3	2	0	0	1	2
2				2	3				3	2	1	0	0
3	2				2	3				3	2	0	0

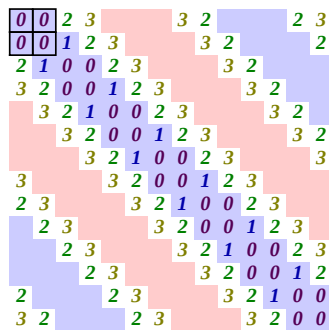
► s_i adds or removes boxes with residue i (in contiguous groups).

Example in $\widetilde{D}_5/D_5$: $s_1 s_2 s_3 s_4 s_5 s_3 s_2 s_0$.

Residue Structure in W/W

In other types, residues involving $\{n-1, n\}$ and $\{0, 1\}$ depend on λ .

In type $\tilde{D}/D$, there are both **escalators** and **descalators**.



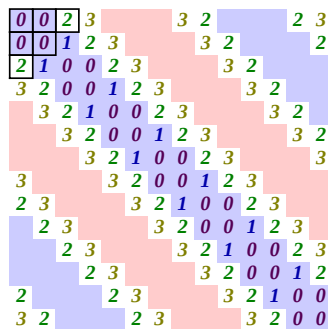
- s_j adds or removes boxes with residue i (in contiguous groups).

Example in $\tilde{D}_5/D_5$: $s_1 s_2 s_3 s_4 s_5 s_3 s_2 s_0$.

Residue Structure in $\widetilde{W}/W$

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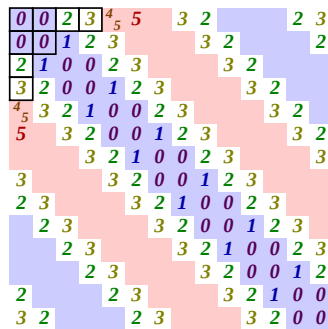
► s_i adds or removes boxes with residue i (in contiguous groups).

Example in $\widetilde{D}_5/D_5$: $s_1 s_2 s_3 s_4 s_5 s_3 s_2 s_0$.

Residue Structure in $\tilde{W}/W$

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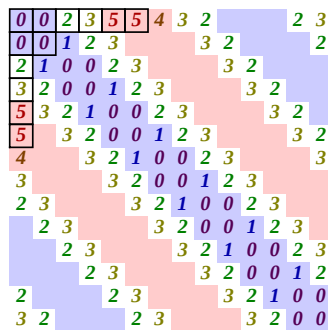
- s_j adds or removes boxes with residue i (in contiguous groups).

Example in $\tilde{D}_5/D_5$: $s_1 s_2 s_3 s_4 s_5 s_3 s_2 s_0$.

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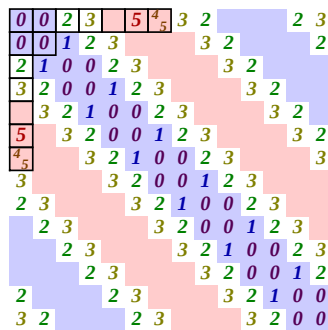
► s_i adds or removes boxes with residue i (in contiguous groups).

Example in $\widetilde{D}_5/D_5$: $s_1 s_2 s_3 s_4 s_5 s_3 s_2 s_0$.

Residue Structure in W/W

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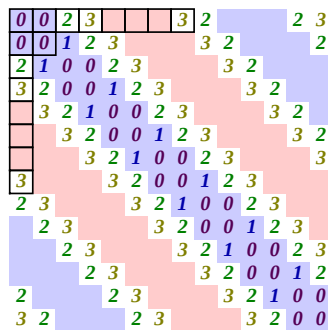
- s_j adds or removes boxes with residue i (in contiguous groups).

Example in $\tilde{D}_5/D_5$: $s_1 s_2 s_3 s_4 s_5 s_3 s_2 s_0$.

Residue Structure in $\widetilde{W}/W$

In other types, residues involving $\{n-1, n\}$ and $\{0, 1\}$ depend on λ .

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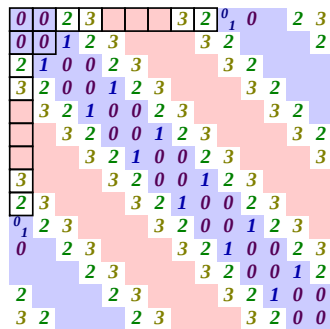
► s_i adds or removes boxes with residue i (in contiguous groups).

Example in $\widetilde{D}_5/D_5$: $s_1 s_2 s_3 s_4 s_5 s_3 s_2 s_0$.

Residue Structure in W/W

In other types, residues involving $\{n-1, n\}$ and $\{0, 1\}$ depend on λ .

In type $\tilde{D}/D$, there are both **escalators** and **descalators**.



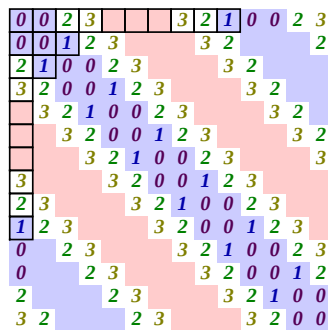
- s_i adds or removes boxes with residue i (in contiguous groups).

Example in $\tilde{D}_5/D_5$: $s_1 s_2 s_3 s_4 s_5 s_3 s_2 s_0$.

Residue Structure in $\widetilde{W}/W$

In other types, residues involving $\{n-1, n\}$ and $\{0, 1\}$ depend on λ .

In type $\widetilde{D}/D$, there are both **escalators** and **descalators**.



► s_i adds or removes boxes with residue i (in contiguous groups).

Example in $\widetilde{D}_5/D_5$: $s_1 s_2 s_3 s_4 s_5 s_3 s_2 s_0$.

Properties of abaci and cores for $\widetilde{W}/W$

Theorem. Minimal length coset representatives in $\widetilde{W}/W$ are in bijection with the following sets of abaci and self-conjugate partitions:

Theorem. The residues in the partitions have the following structure:

Type	Abaci	Partitions	Residues
$\widetilde{C}/C$	all abaci	all self-conj $(2n)$ -cores	fixed
$\widetilde{B}/B$	even abaci	even self-conj $(2n)$ -cores	fixed w/descalators
$\widetilde{B}/D$	all abaci	all self-conj $(2n)$ -cores	fixed w/escalators
$\widetilde{D}/D$	even abaci	even self-conj $(2n)$ -cores	fixed w/descalators and escalators

Canonical reduced expression for $\widetilde{w}/w$

Peel a core to obtain a *canonical reduced expression* for w^0 .

Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0$$

Canonical reduced expression for $\widetilde{w}/w$

Peel a core to obtain a *canonical reduced expression* for w^0 .

Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1$$

Canonical reduced expression for $\widetilde{w}/w$

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0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0$$

Canonical reduced expression for $\widetilde{W}/W$

Peel a core to obtain a *canonical reduced expression* for w^0 .

Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0 s_3$$

Canonical reduced expression for $\widetilde{W}/W$

Peel a core to obtain a *canonical reduced expression* for w^0 .

Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0 s_3 s_2$$

Canonical reduced expression for $\widetilde{w}/w$

Peel a core to obtain a *canonical reduced expression* for w^0 .

Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0 s_3 s_2 s_1$$

Canonical reduced expression for $\widetilde{W}/W$

Peel a core to obtain a *canonical reduced expression* for w^0 .

Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

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0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0 s_3 s_2 s_1 s_0 s_2$$

Canonical reduced expression for $\widetilde{W}/W$

Peel a core to obtain a *canonical reduced expression* for w^0 .

Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0 s_3 s_2 s_1 s_0 s_2 s_3$$

Canonical reduced expression for $\widetilde{W}/W$

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Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

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0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1$$

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Peel a core to obtain a *canonical reduced expression* for w^0 .

Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1 s_0$$

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Peel a core to obtain a *canonical reduced expression* for w^0 .

Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1 s_0 s_2$$

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Peel a core to obtain a *canonical reduced expression* for w^0 .

Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1 s_0 s_2 s_3$$

Canonical reduced expression for $\widetilde{W}/W$

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Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1 s_0 s_2 s_3 s_2$$

Canonical reduced expression for $\widetilde{W}/W$

Peel a core to obtain a *canonical reduced expression* for w^0 .

Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1$$

Canonical reduced expression for $\widetilde{W}/W$

Peel a core to obtain a *canonical reduced expression* for w^0 .

Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1 s_0$$

Canonical reduced expression for $\widetilde{W}/W$

Peel a core to obtain a *canonical reduced expression* for w^0 .

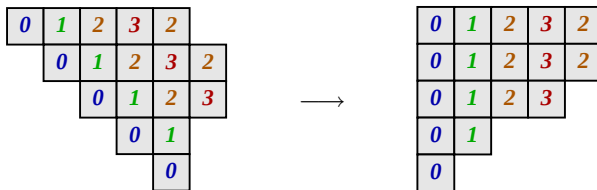
Remove boxes from the center and record the residues at each step.

0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0
2	3	2	1	0	1	2	3	2	1
1	2	3	2	1	0	1	2	3	2
0	1	2	3	2	1	0	1	2	3
1	0	1	2	3	2	1	0	1	2
2	1	0	1	2	3	2	1	0	1
3	2	1	0	1	2	3	2	1	0

$$\mathcal{R}(\lambda) = s_0 s_1 s_0 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1 s_0 s_2 s_3 s_2 s_1 s_0$$

Bounded partitions for $\widetilde{W}/W$

Left-justifying this **upper diagram** gives a **bounded partition** (satisfying type-dependent conditions).

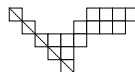
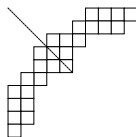
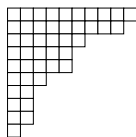


Bounded partitions:

- ▶ Encode a reduced expression for the element
- ▶ Have Coxeter length number of boxes
- ▶ Have appeared in crystal basis theory (as Young walls), in work of Eriksson–Eriksson, and in work of Billey–Mitchell (as affine colored partitions)

Lapointe–Morse-like bijection for bounded partitions

- ▶ Remove all boxes of λ with hook length $\geq 2n$
- ▶ Reinsert the boxes on the main diagonal, remove those below.
- ▶ Left-justify remaining boxes to diagonal.
- ▶ (When not $\tilde{C}/C$, remove boxes on main and/or n -th diagonal.)
- ▶ **Result:** Upper diagram.



$$\lambda = (10, 9, 6, 5, 5, 3, 2, 2, 2, 1)$$

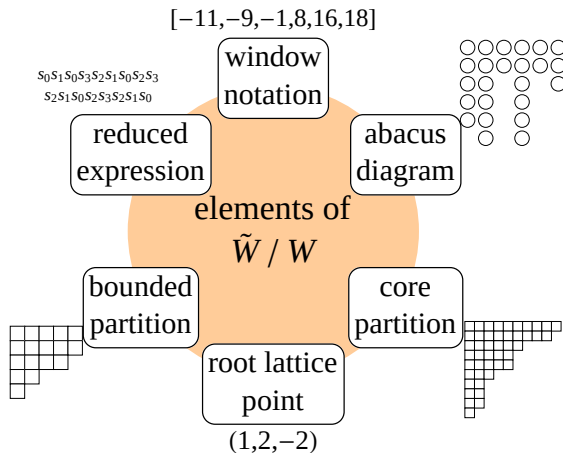
$$\beta = (5, 5, 4, 2, 1)$$

Conditions on bounded partitions

Theorem. We have a bijection of $\widetilde{W}/W$ with these bounded partitions:

Type	Bounded partition structure
$\widetilde{C}/C$	parts $\leq 2n$, where $1, \dots, n$ occur at most once.
$\widetilde{B}/B$	parts $\leq 2n - 1$, where $1, \dots, n - 1$ occur at most once.
$\widetilde{B}/D$	parts $\leq 2n - 1$, where $1, \dots, n - 1$ occur at most once, and one n part may be starred.
$\widetilde{D}/D$	parts $\leq 2n - 2$, where $1, \dots, n - 2$ occur at most once, and one $n - 1$ part may be starred.

Combinatorial interpretations in $\widetilde{W}/W$



Future Work

- ▶ More combinatorial interpretations for $\widetilde{W}/W$
 - ▶ Learn more about the alcove model
 - ▶ What do you know?
- ▶ Fully commutative elements in types $\widetilde{B}$, $\widetilde{C}$, and $\widetilde{D}$
 - ▶ Investigation in $\widetilde{A}$ required combinatorial interpretations ✓
 - ▶ Find a 321-avoiding characterization?
- ▶ Self-conjugate core partitions
 - ▶ Related to $\widetilde{C}_n/C_n$.
 - ▶ Related to the alternating group.

Thank you!

Slides available: people.qc.cuny.edu/chanusa > Talks

Interact: people.qc.cuny.edu/chanusa > Animations



Christopher R. H. Hanusa and Brant C. Jones.

Abacus models for parabolic quotients of affine Coxeter groups
[arXiv:1105.5333](https://arxiv.org/abs/1105.5333)



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The enumeration of fully commutative affine permutations
European Journal of Combinatorics. Vol 31, 1342–1359. (2010)



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